

Solution Requirements

Date	12 August 2026
Name	Chahat Chouhan
Team ID	Individual Project
Project Name	AI-Brand-Voice-Generator
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely authenticate via email/password or OAuth 2.0 (Google/GitHub) with session token verification.	High
2	Authorization levels	Role-Based Access Control (RBAC): Admin (manage all users/system settings), Marketer (create/edit brand profiles and generate content), Viewer (read-only history).	High

3	External interfaces	Connects to Google Gemini API for NLP extraction and content generation, and SQLite/PostgreSQL database for data persistence.	High
4	Transactions processing	Handles real-time API request execution, brand profile creation workflows, and content generation query logs with retry mechanisms on network failure.	High
5	Reporting	Interactive dashboard displaying key metrics (tokens used, generation history, active brand profiles) and downloadable export options (JSON/PDF).	Medium
6	Business rules, process	Text sample uploads for analysis must be between 100 and 10,000 words. Content prompts must adhere to brand voice guidelines stored in the DB before dispatch.	High
7	Compliance to laws or regulations	Adheres to GDPR and CCPA standards for user data privacy, with automated data	Medium

		deletion mechanisms upon user account deletion request.	
8	Other	Multi-language support for international brand voice analysis and auto-saving of active session drafts.	Low

Step 2: Non-Functional Requirements (NFR)

S. No.	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Web UI response latency and initial page load times under standard network conditions.	Page loads in < 2 seconds; API response time < 3 seconds.
2	Scalability	Ability to handle scaling user traffic and concurrent background content generation tasks.	Supports up to 10,000 concurrent active users with horizontal scaling.
3	Security & Data Privacy	Data protection, transport encryption, and credential handling compliance.	End-to-end AES-256 encryption for data at rest and TLS 1.3 in transit.
4	Reliability &	System uptime	99.9% uptime per

	Availability	operational guarantee and fault tolerance capability.	month with automated failover handling.
5	Usability & Accessibility	Clean UI design, responsive layout, and universal accessibility compliance across web browsers.	Complies with WCAG 2.1 AA standards; fully responsive on mobile/desktop.
6	Other	System maintainability, code modularity, and automated container deployment support.	Dockerized container setup with > 80% test code coverage.